Oligopoly and Game Theory

Module 7



Management 405

Prof. Nico Voigtländer · UCLA Anderson

1

Exam-Specific Announcements

- Final Exam: Sat 12/12, 9:00 am – 12:00 pm
  - 3 hours for the exam, then 15 minutes to upload your PDF
  - Same logistics as the Midterm Exam (see the next slides)
- Practice Final Exam on the class website
  - Similar to the actual final
- More practice: online quizzes (see the class website)

2

Final Exam: Structure

- Covers material from the full quarter: Modules 1 – 8
  - All modules relevant, more weight on post-midterm topics
- 40% Multiple Choice: about 20 questions, 2 points each
  - Directly on BruinLearn; only one answer correct, no partial credit
- 60% Problem Solving: 3 – 4 problems
  - Type final answers on BruinLearn; upload ONE PDF with your detailed steps
  - Show your work for full / partial credit, even if you get stuck or run out of time
- Extra assumptions: write them down and email the TA after the exam

3

Final Exam: Logistics

- Same logistics as the Midterm Exam
  - Respondus LockDown Browser + webcam; show your workspace at the start
  - Anything on paper is allowed; own **summary ("cheat") sheets** strongly recommended
  - TA on Zoom during the exam: join from your phone, as for the midterm
  - No other devices, internet, or AI; calculator is built into the browser
  - After the exam: webcam photos, then upload ONE PDF within 15 minutes
- Different from the midterm
  - One fixed time slot for everyone: Sat 12/12, 9:00 am – 12:00 pm
  - TA will be available via Zoom (same logistics as during Midterm)

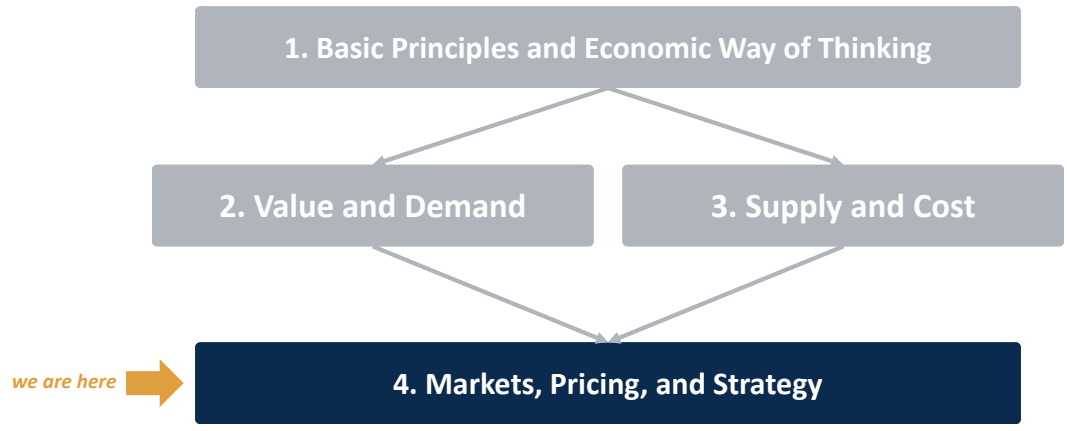
4

Further Exam-Related Announcements

- Review session with Nico: Sun 12/6, time TBA
  - Send me topics you'd like to be covered!
  - Additional office hours on demand
- TA review sessions and practice sessions: see email announcements
  - More review videos by the TA during the exam-prep weeks
  - «Last-minute» review with the TA: Fri 12/11, time TBA

5

Agenda for the Class



6

Outline of Module 7

1. **Introduction to Oligopoly and Game Theory** — Video 1
   What makes a market an oligopoly, and why firms must think strategically
2. **Collusion and Cartels** — Video 2
   Why firms are tempted to collude, and how cartels can break down
3. **Cournot Competition** — Video 3
   Competing by choosing quantities: capacity decisions in oligopoly
4. **Bertrand Competition** — Video 4
   Competing by setting prices, and why it can compete profits away
5. **Bertrand Competition with Differentiated Goods** — Video 5
   How differentiation softens price competition
6. **Game Theory** — In class
   - 6a **Strategic Thinking: Key Concepts** — In class
     Best responses, dominant strategies, and Nash equilibrium
   - 6b **Prisoner's Dilemma, Game of Chicken…** — In class
     Classic games and what they teach about business strategy

7

Module 7
In-Class Part

8

Some Applications of the Material Covered in the Videos

9

Recall: Collusion and Cartels

- A cartel seeks to raise market prices by reducing supply
- At the inflated cartel price, each firm would like to produce more than its reduced amount
  → Incentives to deviate
- How do cartels coordinate to avoid deviation?



10

Collusion: The Case of ADM

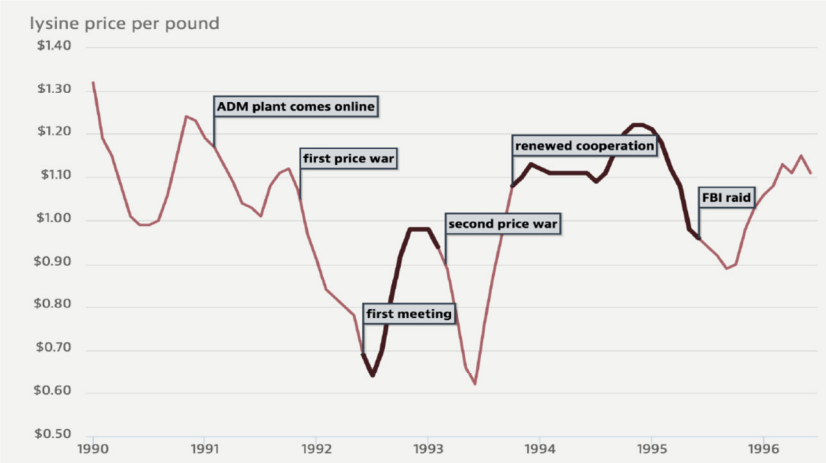
- Archer Daniels Midland (ADM) involved in organized effort to raise the price of the animal feed additive lysine
- Unique case with FBI mole
  - Taped price-fixing meetings
- Original FBI footage on course webpage (under Module 7 In-Class Content)
- Podcast, Movie, Related Podcast: The Poop Cartel



Matt Damon in "The Informant!" (2009)

11

The ADM Price-Fixing Scandal



Source: American Economic Association, "Conspiracies against the public" (aeaweb.org)

12

For the Rest of Module 7: Competition, Not Collusion

No collusion among competitors

→ **Firms compete strategically ("games")**



13

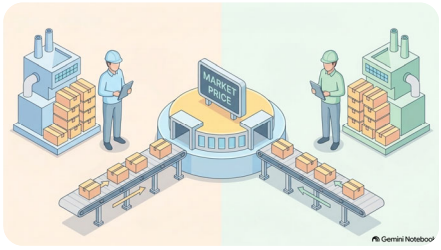
Two Models of Oligopoly Competition

Cournot

- The rival's supply is taken as given
- Firm reacts by adjusting its own quantity accordingly
- Price is determined in the market

Bertrand

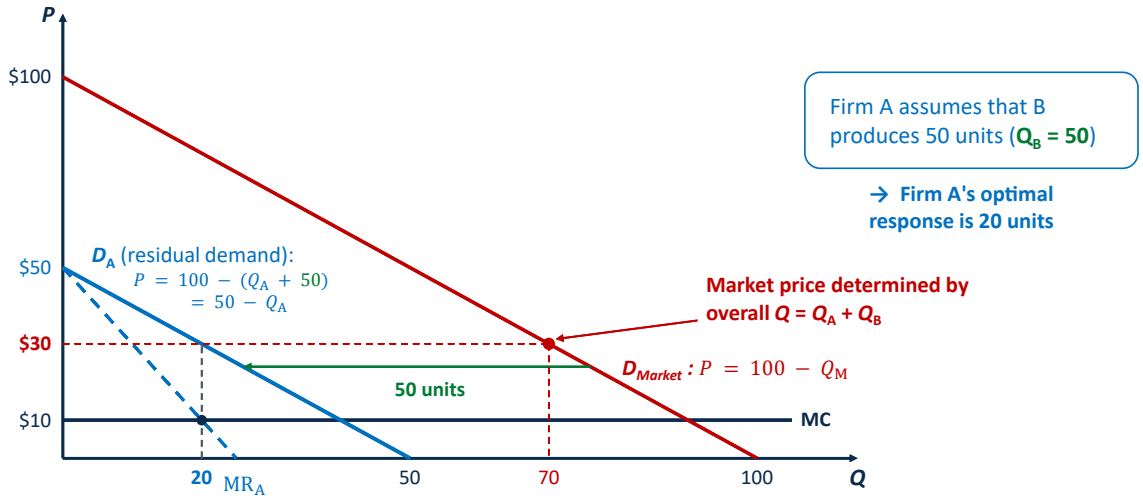
- The rival's price is taken as given
- Firm reacts by adjusting its own price (underbidding, if feasible)
- Firms then supply the quantity that is demanded at the price they set





14

Cournot Competition: Reaction Function (I)



Firm A assumes that B produces 50 units (Q_B = 50)

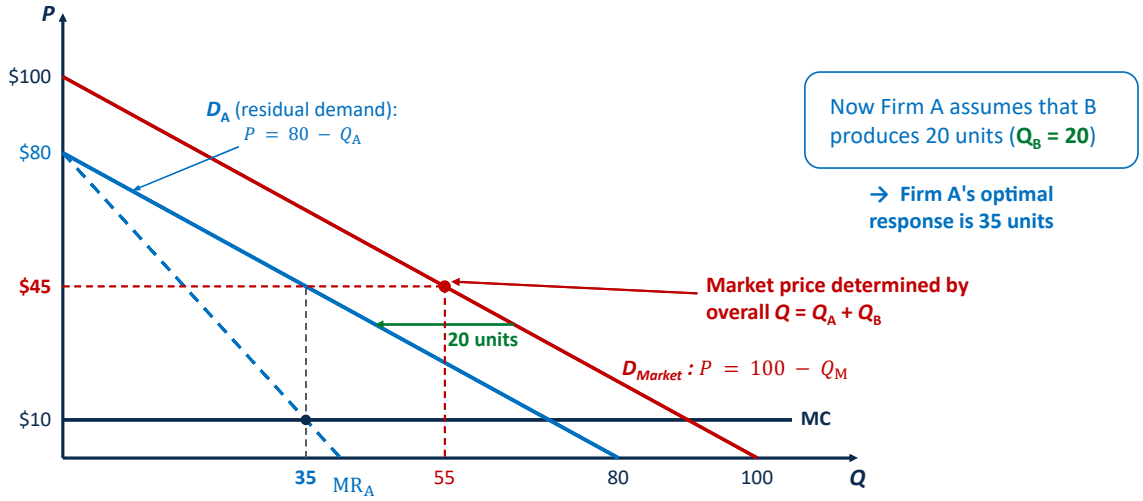
→ Firm A's optimal response is 20 units

Market price determined by overall $Q = Q_A + Q_B$

$D_{Market}: P = 100 - Q_M$

15

Cournot Competition: Reaction Function (II)



Now Firm A assumes that B produces 20 units (Q_B = 20)

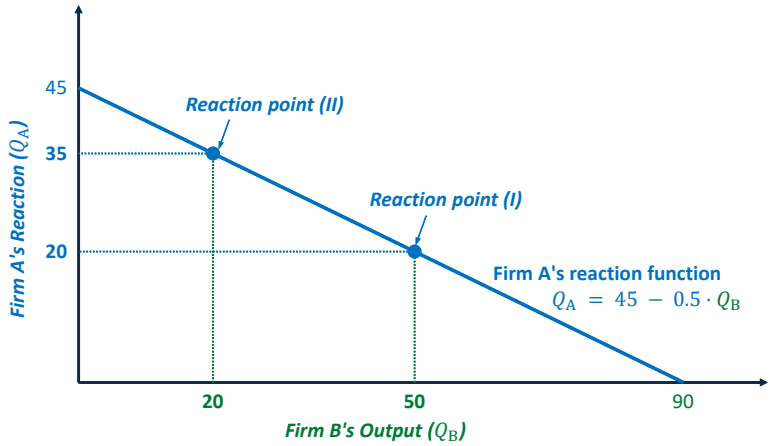
→ Firm A's optimal response is 35 units

Market price determined by overall $Q = Q_A + Q_B$

$D_{Market}: P = 100 - Q_M$

16

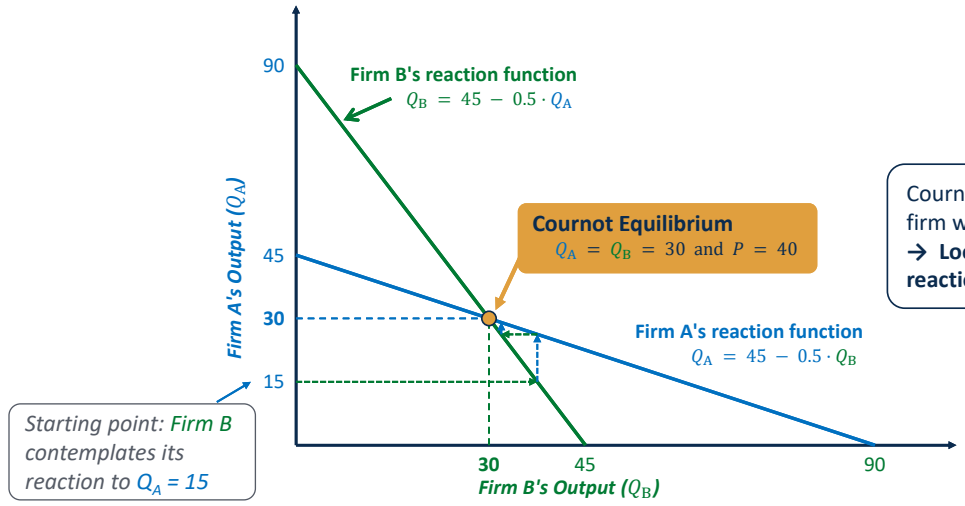
Firm A's Reaction Function



Practice Video "Cournot Competition Math"

17

Cournot Equilibrium



Cournot equilibrium: neither firm would like to deviate
→ **Look for intersection of reaction functions**

Starting point: Firm B contemplates its reaction to $Q_A = 15$

18

Cournot: Math – Also Relevant for the Exam

Practice Video "Cournot Competition Math"

More general case with different MC:

Practice Video "Oligopoly with different MC"

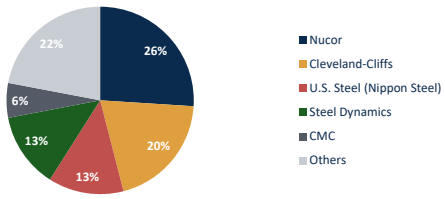
Examples for Cournot Competition

- Oil
  - However: Includes a cartel (OPEC)



- Steel producers

US crude steel production by producer, 2024



Source: worldsteel, company reports. U.S. Steel acquired by Nippon Steel, June 2025

Bertrand Competition (with Identical Goods)

Assumptions:

- Firms make identical products
- Firms compete by choosing the price at which they sell their products



Choose Your Concrete Price

- Two competitors, identical good: Concrete
- Same marginal cost: MC = $120 per cubic yard
- You and your competitor submit a sealed price bid for a huge project
- What's your price bid?





Poll Break

What price bid would you (being the producer of concrete) submit?

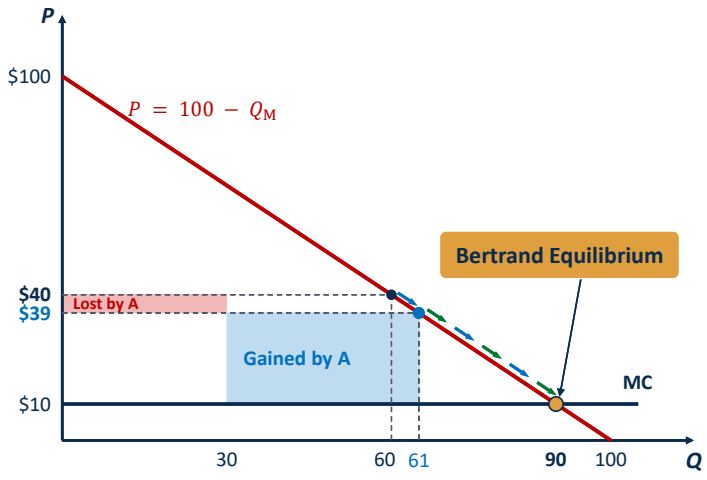
$>300	0%
$150-200	0%
$140-150	0%
$130-140	0%
$125-130	0%
$120-125	0%
$120	0%
$100-120	0%
<$100	0%

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23

Module 7 · Video 4 · Bertrand Competition

Bertrand Competition



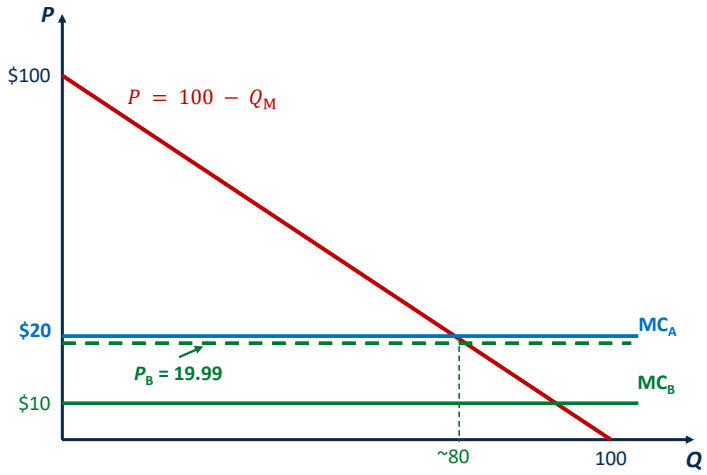
Suppose Firms A and B start at a price of $40, where each supplies Q=30
A believes that B will maintain its price. What will A do? How will B respond?

→ A and B will continue to lower their price until they reach P = MC

24

Module 7 · Video 4 · Bertrand Competition

Bertrand Competition with Different MC



Now Firm A has $MC_A = 20$
Firm B still has $MC_B = 10$

Firm B (with lower MC_B) will charge a price just under MC_A. This pushes **Firm A** out of the market.

25

Module 7 · Video 5 · Bertrand Competition with Differentiated Goods

Bertrand Competition (with Differentiated Goods)

Assumptions:

- Firms do not sell identical products. They sell differentiated products
- Each firm faces its own demand function
- Each firm chooses its price, responding to the price of the other firm



26

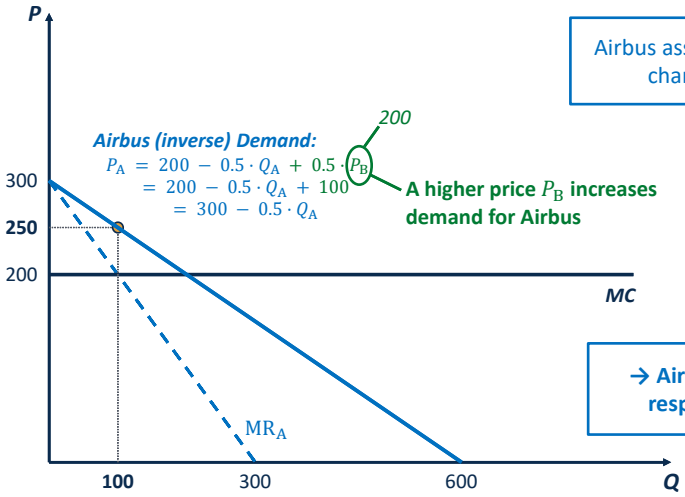
Reaction Function: Airbus' Reaction to Boeing's Price (I)

Note: Airbus Demand:

$Q_A = 400 - 2P_A + P_B$

Derive inverse Airbus Demand:

$\rightarrow P_A = 200 - 0.5 \cdot Q_A + 0.5 \cdot P_B$



Airbus (inverse) Demand:

$P_A = 200 - 0.5 \cdot Q_A + 0.5 \cdot P_B$
$= 200 - 0.5 \cdot Q_A + 100$
$= 300 - 0.5 \cdot Q_A$

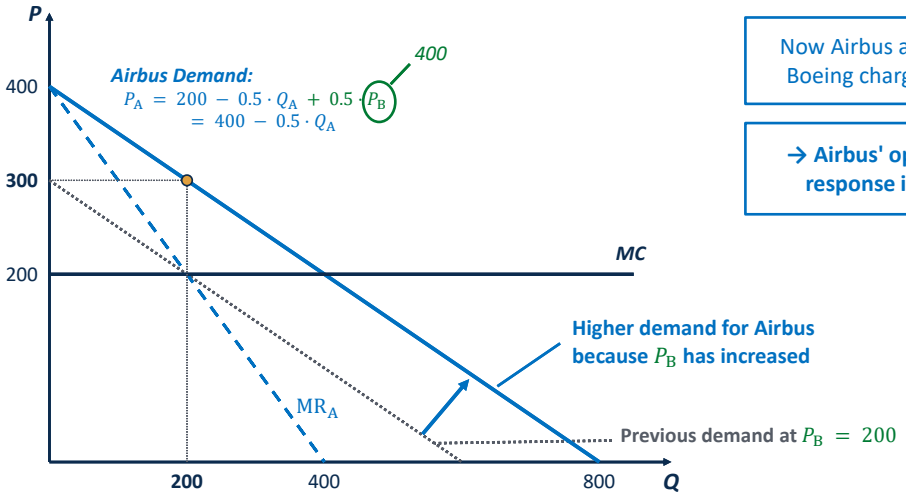
200

A higher price P_B increases demand for Airbus

Airbus assumes that Boeing charges $P_B = 200$

→ Airbus' optimal price response is $P_A = 250$

Reaction Function: Airbus' Reaction to Boeing's Price (II)



Airbus Demand:

$P_A = 200 - 0.5 \cdot Q_A + 0.5 \cdot P_B$
$= 400 - 0.5 \cdot Q_A$

400

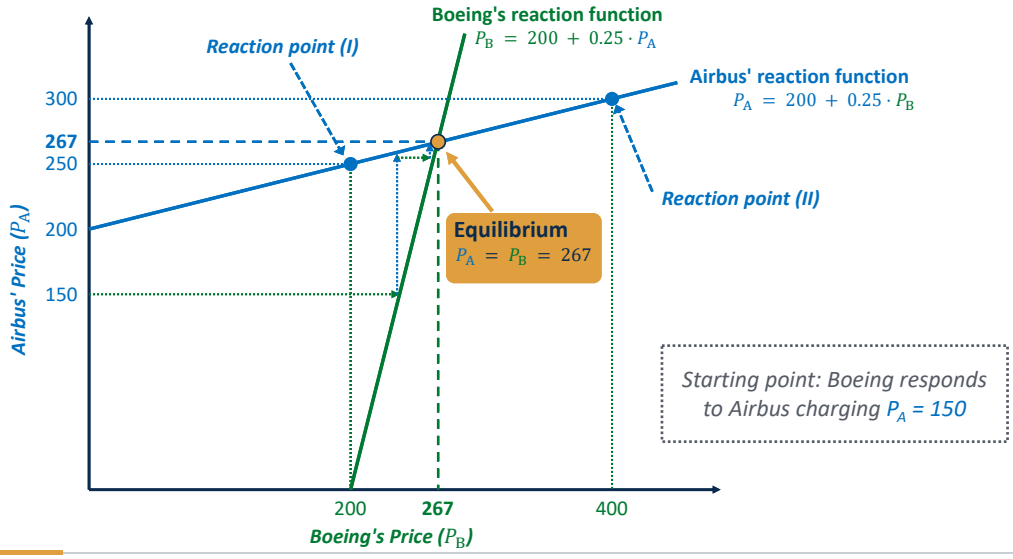
Now Airbus assumes that Boeing charges $P_B = 400$

→ Airbus' optimal price response is $P_A = 300$

Higher demand for Airbus because P_B has increased

Previous demand at $P_B = 200$

Equilibrium Prices



Play This Game on the Class Website

- Under Module 7
- You can choose Boeing's price and observe Airbus's reaction (price)
  - You also see the profits of both companies
- You can also change the demand functions (make payoffs asymmetric)

Class Website → Module 7

Oligopoly: Concluding Discussion

- If you could pick one type of Oligopoly competition for your firm, which would you choose? Why?
- In which oligopoly case can you improve your product?
  - What is the effect of improving your product?

Discussion

31

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6a	**Strategic Thinking: Key Concepts** Best responses, dominant strategies, and Nash equilibrium	In class
6b	Prisoner's Dilemma, Game of Chicken…	In class

32

What Is Game Theory?

- Branch of economics concerned with interdependent situations
  - The outcome of a participant's choice depends critically on the actions of other participants
- Game theory has transformed not only economics, but also the study of war, of business strategy, and of biological systems
- Father of Game Theory: John Nash
  - 1994 Nobel Prize in Economics
  - Inspired the movie "A Beautiful Mind"



John Nash (1928–2015)





A Warm-Up Game

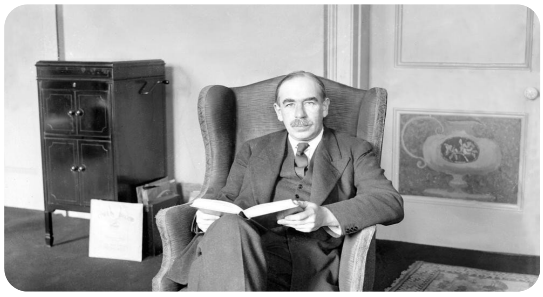
- I brought 5 yellow and 5 red cards
- But this is not about soccer…
- Instead:
  - I distribute the 5 yellow cards to students
  - Suppose I get $100 from the Dean for each red-yellow pair that I take back to her after class
  - You can 'sell' your yellow card back to me
  - What's your asking price for your yellow card?



World Cup Final, 2006

Strategic Thinking: The Beauty Contest

- Prominent game in newspapers in the 1930s:
  - Get a reward if you select the face that is the most popular among others (not the one that you find most attractive)
- John Maynard Keynes (1936) used it as analogy

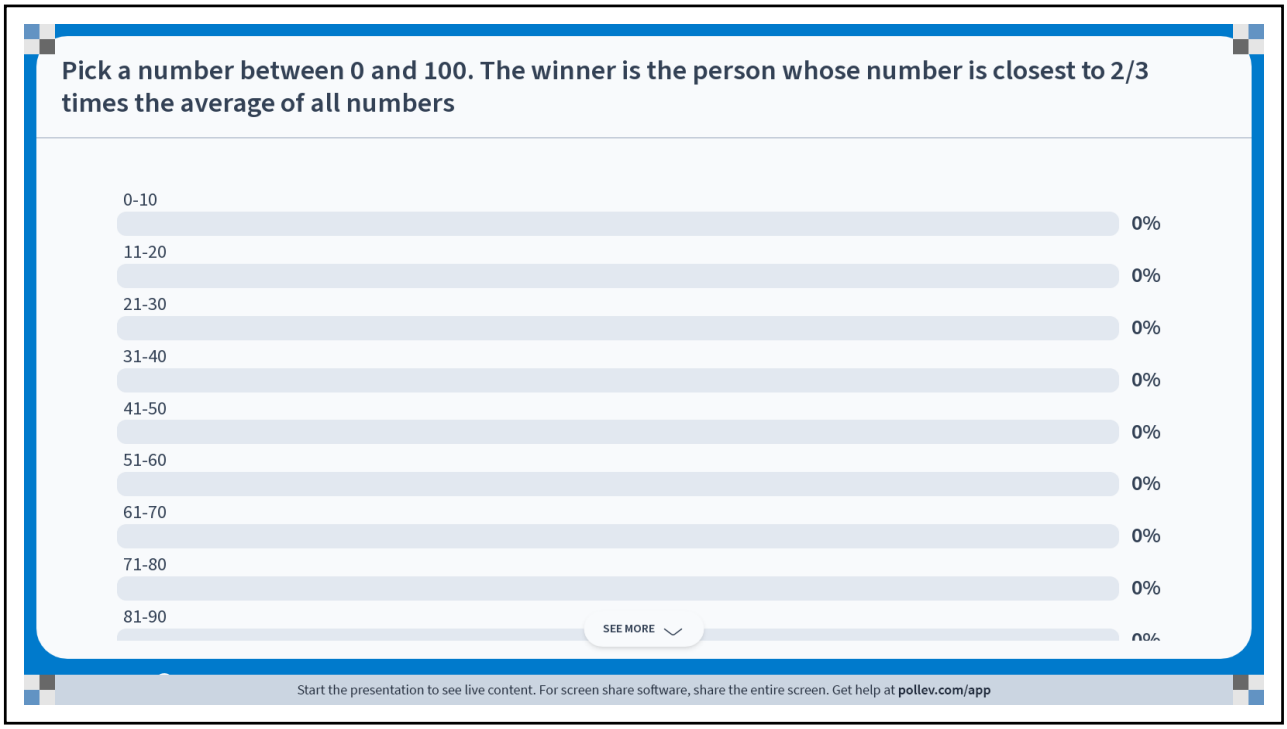


John Maynard Keynes

Strategic Thinking: Numerical Beauty Contest

- Numerical application:
  - All participants are asked to pick a number between 0 and 100
  - Winner: the number that is closest to 2/3 of the average of all numbers submitted

Poll Break



37

Key Concepts of Game Theory

- Strategic behavior by all participants
  - Optimization: maximize *individual* payoff
- To study strategic behavior:
  - **Best response**: optimal decision given what the other player does
  - **Dominant strategy**: optimal action no matter what the other player does
  - **Nash equilibrium**: «intersection of best responses»
    - Position from which neither player would find it advantageous to deviate (e.g., Cournot equilibrium)

38

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6b	**Prisoner's Dilemma, Game of Chicken…** Classic games and what they teach about business strategy	In class

Prisoner's Dilemma: Setup and Payoffs

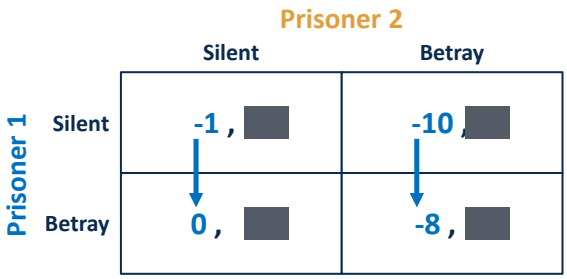
Prisoner 1 \ Prisoner 2	Silent	Betray
Silent	-1 , -1	-10 , 0
Betray	0 , -10	-8 , -8

Payoffs to (Prisoner 1, Prisoner 2)

Interpretation of payoffs:
Negative values, because they indicate years in prison



Prisoner's Dilemma: Strategy of Prisoner 1



*Payoffs to (**Prisoner 1**, **Prisoner 2**)*

Interpretation of arrows:
Prisoner 1 picks his best response to Prisoner 2's (hypothetical) choices

41

Prisoner's Dilemma: Strategy of Prisoner 2



*Payoffs to (**Prisoner 1**, **Prisoner 2**)*

Interpretation of arrows:
Prisoner 2 picks his best response to Prisoner 1's (hypothetical) choices

42

Prisoner's Dilemma: Equilibrium

Prisoner 1 \ Prisoner 2	Silent	Betray
Silent	-1 , -1	-10 , 0
Betray	0 , -10	-8 , -8

Payoffs to (Prisoner 1, Prisoner 2)

Nash Equilibrium

Both players' *dominant strategy* is to betray, so individual optimization produces a "socially" sub-optimal outcome

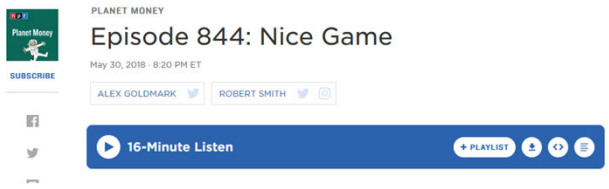
Ways to Solve the Dilemma

- Communication:
  - What would a lawyer do?
- External solutions:
  - What would the Mafia do?
  - Commitment
  - Moral rules (guilt/"conscience")
- Repetition
  - Podcast (next slide)



Podcast on Prisoner's Dilemma

- What's the experiment that Bob Axelrod ran in the 1980s?
- What was the winning strategy?



Fair Play vs. "Dirty" Play

Team 1 \ Team 2	Play fair	Play dirty
Play fair	0.5 , 0.5	0.3 , 0.7
Play dirty	0.7 , 0.3	0.5 , 0.5

Payoffs to (Team 1, Team 2)

Nash Equilibrium



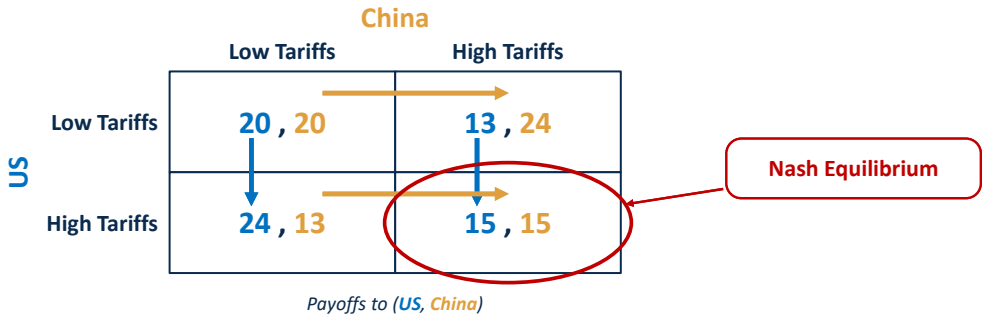
World Cup Final, 2010

Interpretation of payoffs:
Teams' odds of winning

Each team's dominant strategy is to play dirty

Game Theory and Trade Wars

In a trade war, each country's dominant strategy is to set high tariffs



Interpretation of payoffs:
Countries' real GDP ($ trillion, price-adjusted)

2025–26 tariff war: US average tariff on Chinese goods ≈ 48%; China's on US goods ≈ 32% (PIIE)

General Lesson: Collective Action Problem

- Acting selfishly can lead to socially suboptimal outcomes
  - Also holds in situations with more than 2 players
  - Known as "tragedy of the commons"

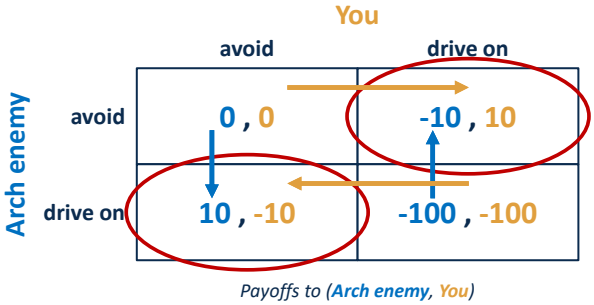


- Stark contrast with "invisible hand" principle (Adam Smith)

The Game of Chicken

You and your arch-enemy drive towards each other on a narrow street

[Similar game in "Rebel Without a Cause"]



Payoffs to (Arch enemy, You)

2 Nash Equilibria

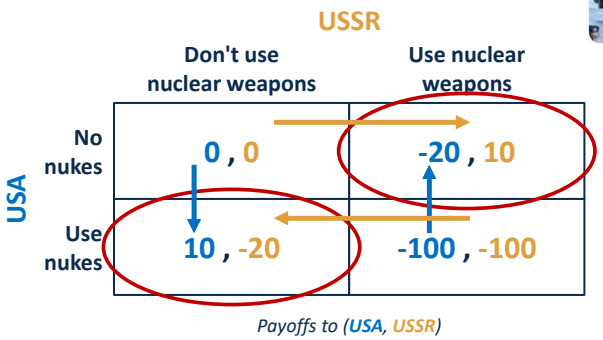
- Real world examples:
  - Entry into a market that can only sustain one firm profitably (natural monopoly)
  - The Cold War (next slide)

The Cold War and M.A.D.

- Why did none of the two super-powers use nukes?
  - Mutually Assured Destruction (M.A.D.)







Payoffs to (USA, USSR)

2 Nash Equilibria

Group Work

- How does the payoff table change under M.A.D.?
  - "Doomsday device" (also explained by Dr. Strangelove)



Dr. Strangelove

		USSR: Don't use nuclear weapons	USSR: Use nuclear weapons
USA	No nukes		
USA	Use nukes		

Payoffs to (USA, USSR)

The world came very close to destruction: **Documentary**

Discussion

Commitment and Strategy

- Note that M.A.D. represents a commitment (to retaliate)

		USSR: Don't use nuclear weapons	USSR: Use nuclear weapons
USA	No nukes		
USA	Use nukes		

Payoffs to (USA, USSR)

- (Much) more on commitment in your Strategy class!
  - Crucial: Is the player's "commitment" credible?

Tickets →

Penalty Kicks: Find the (Pure-Strategy) Nash Equilibrium

- Note:
  - Pure Strategy: Each player picks either "Left" or "Right" (as in all games we've seen thus far)
  - Mixed Strategy: Players can also assign probabilities with which they pick Left or Right (not relevant for our class)

Kicker \ Goalie	Left	Right
Left	-1 , 1	1 , -1
Right	1 , -1	-1 , 1

*Payoffs to (**Kicker**, **Goalie**)*

Poll Break

53

Penalties: Find the Pure Strategy Nash Equilibrium

(A) Left, Left 0%

(B) Left, Right 0%

(C) Right, Left 0%

(D) Right, Right 0%

(E) Every option is a Nash equilibrium 0%

(F) There is no (pure strategy) Nash equilibrium 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

54

Solution

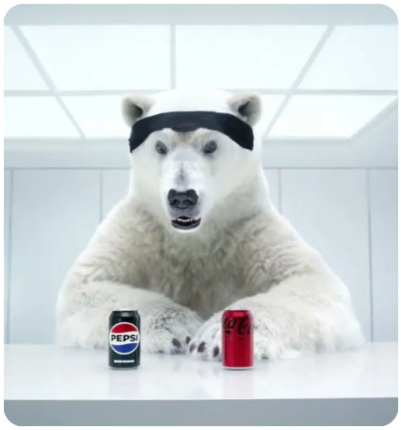
Poll Break

55

Find the (Pure-Strategy) Nash Equilibrium

Coke \ Pepsi	Advertise	Don't Advertise
Advertise	150 , 150	450 , -75
Don't Advertise	-75 , 450	225 , 225

Payoffs to (Coke, Pepsi)



Interpretation of payoffs:
Firms' profits

Poll Break

56

Coke and Pepsi: Find the Pure Strategy Nash Equilibrium

Advertise, Advertise	0%
Advertise, Don't Advertise	0%
Don't Advertise, Advertise	0%
Don't Advertise, Don't Advertise	0%
Every option is a Nash equilibrium	0%
There is no (pure strategy) Nash equilibrium	0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

57

Solution

Poll Break

58

Game Theory on TV: Split or Steal?



"Split or Steal": Payoff Matrix

- We assume that good reputation (when splitting) is worth £5k (while bad reputation is worth £0)
- We round the total jackpot to £100k

		Sarah: Split	Sarah: Steal
Steven	Split		
Steven	Steal		

*Payoffs to (**Steven**, **Sarah**)*

2 Nash Equilibria

Split or Steal: Cheap Talk



61

If I were Sarah I would:

Split 0%

Steal 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

62

If I were Steve I would:

Split 0%

Steal 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at pollev.com/app

63

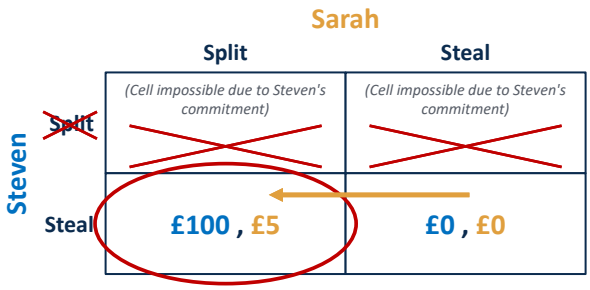
Split or Steal



64

Commitment and Strategy

- If Steven could commit to Steal, what would be Sarah's optimal response?

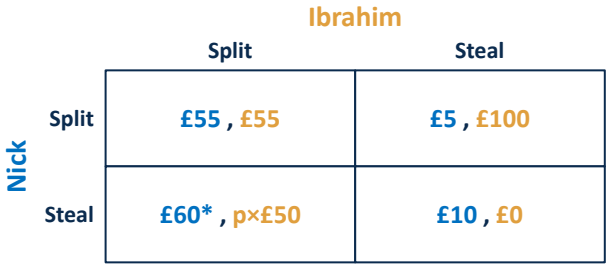


*Payoffs to (**Steven**, **Sarah**)*

- (Much) more on this in your Strategy class!
  - Crucial: Is the player's "commitment" credible?

"Split or Steal": Payoff Matrix

- Nick credibly argues that he will steal. Let's assume he gets "satisfaction" = 10 from stealing and seriously plans to share 50-50
- Good reputation for splitting is still worth 5
- Ibrahim believes that Nick will share winnings with probability $p > 0$



*Payoffs to (**Nick**, **Ibrahim**)*

**Assuming that payoff is shared after stealing. Thus, payoff = 0.5×100 + 10*

Boeing vs. Airbus: Entry Game

- 2 firms: Boeing and Airbus
- One type of product: Ultra-long-range (ULR) aircraft
- Decision: produce or not produce
  - Boeing: incumbent (777-200LR)
  - Airbus: must decide whether to enter (A350 ULR)
- Analyze the situation with and without subsidies





No Subsidy Case

Interpretation of payoffs:
Annual profits from ULR aircraft (in $B)

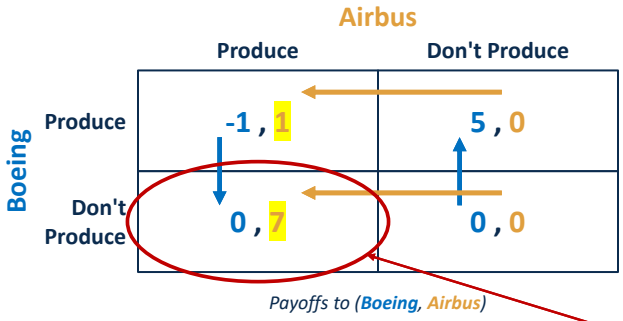
Boeing \ Airbus	Produce	Don't Produce
Produce	-1 , -1	5 , 0
Don't Produce	0 , 5	0 , 0

Payoffs to (Boeing, Airbus)

Boeing is already incumbent. So we're initially in this Nash equilibrium

Subsidy Case

- EU pays $2B to Airbus if they produce

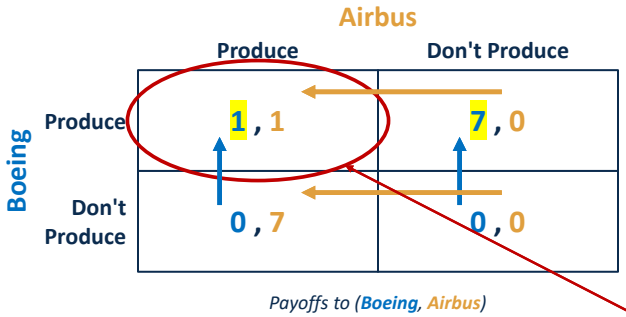


Unique Nash equilibrium in the EU-only subsidy case

69

Subsidy in Both EU and US

- US now also pays $2B to Boeing if they produce



Unique Nash equilibrium in the EU+US subsidy case

70

Recap of Module 7 (For Your Notes)

Oligopoly

- Competition with homogeneous products:
  - Cournot: Firms compete on quantity. Positive profits
    - Equilibrium = intersection of the reaction functions
  - Bertrand: Firms compete on price. Zero profit if both have same MC
    - Price wars → same outcome as perfect competition (if same MC)
- Competition with differentiated products:
  - The two products are imperfect substitutes (differentiated)
  - Bertrand: Each firm sets price for its differentiated product
    - If one firm sets a higher price, the other firm also reacts by raising its price
    - Equilibrium = intersection of the reaction functions

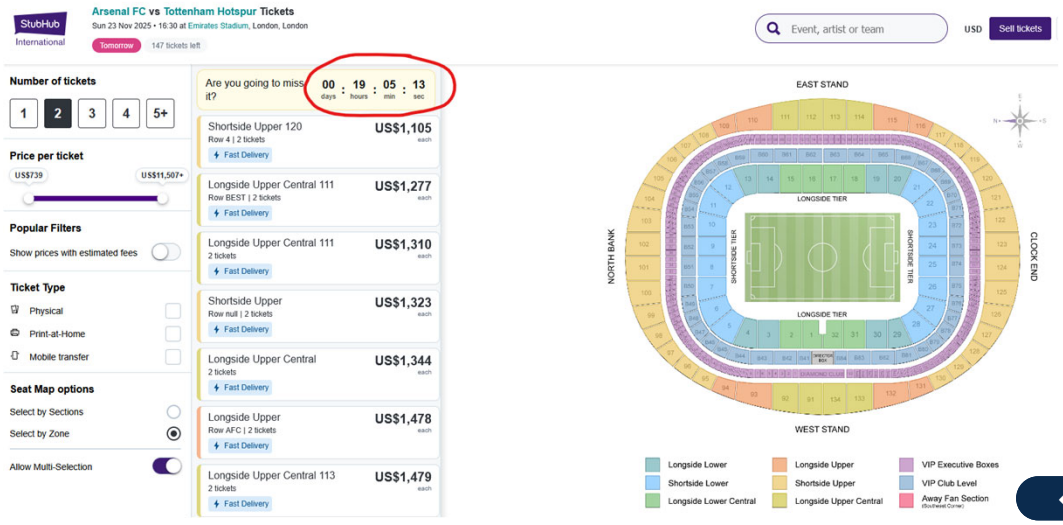
Game Theory

- Strategic thinking: Anticipate other player's reaction
  - Strategy: Optimized reaction to each action by other player
- Use payoff tables
  - Dominant Strategy
  - Nash Equilibrium

BACKUP

Module 7 · Backup

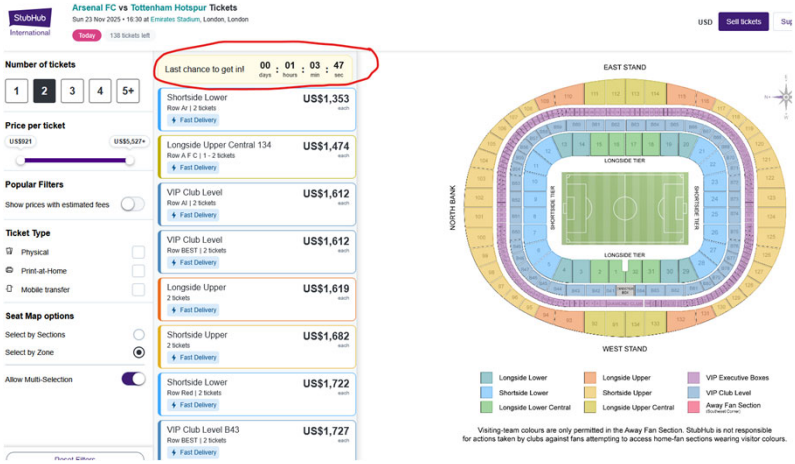
Commitment in Selling Tickets



← Back

Module 7 · Backup

Commitment in Selling Ticket Sales



No sign of falling prices as the game comes closer. Why?

← Back